Morocco Market Stress Index

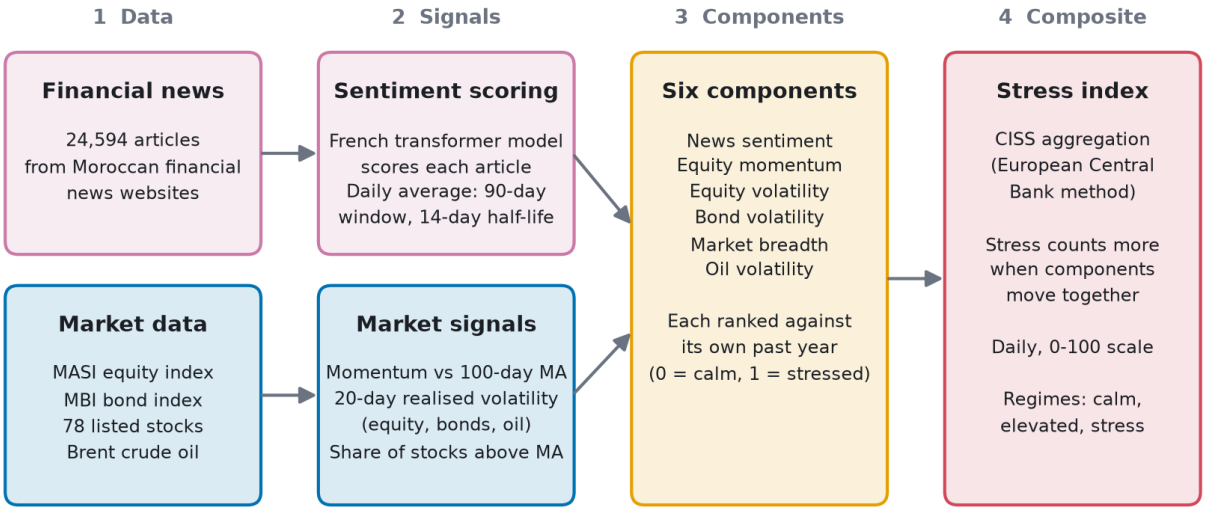
A daily 0-100 gauge of stress on the Casablanca Stock Exchange, built from financial news and market data | 25 Nov 2019 to 17 Jul 2026

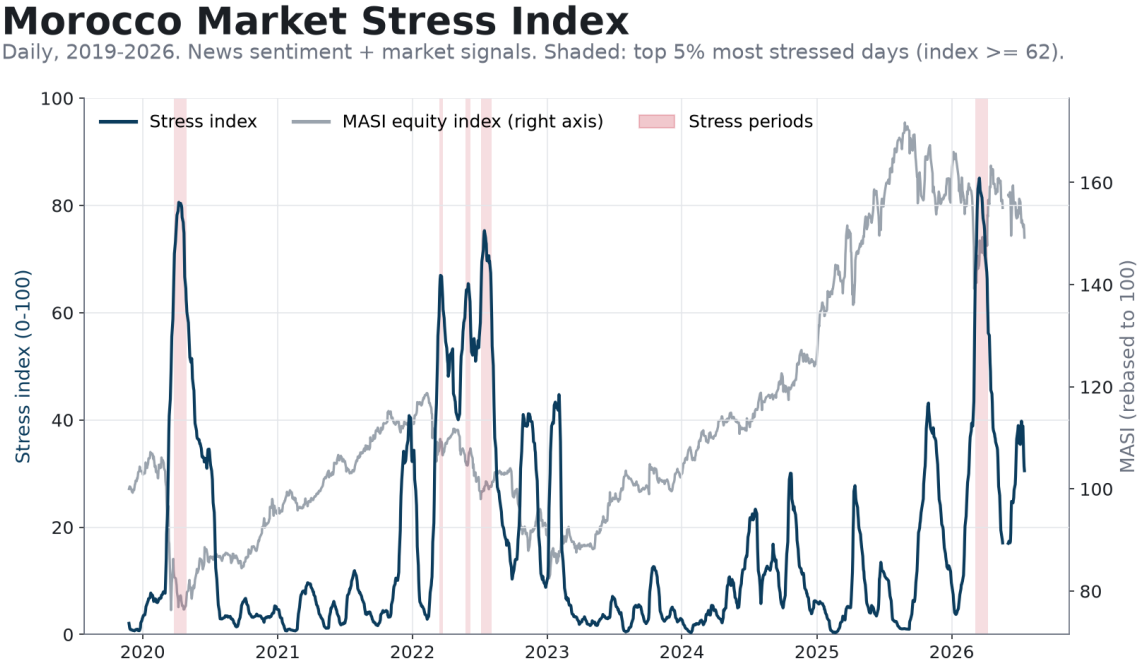
The problem

Investors in Moroccan equities have no widely used daily stress gauge comparable to the VIX for US stocks. Market signals such as volatility react only once prices have moved, while the news flow often turns first. The goal is one daily number that rises when several sources of stress flare up at the same time, built only from information available on each day, so that it can feed allocation and risk decisions.

The method

- **News tone.** 24,594 articles from 2 Moroccan financial news websites are scored by a French language model, from reassuring to alarming. The daily press tone averages the last 90 days of articles, recent ones counting more.
- **Market signals.** MASI trend, volatility of equities (MASI), bonds (MBI) and oil (Brent), and the share of the 78 listed stocks above their recent average.
- **One scale.** Each of the six signals is ranked against its own past year (0 = calmest, 1 = most stressed).
- **Systemic aggregation.** The CISS formula of the European Central Bank combines the six ranks and their correlations: stress that hits several channels at once counts much more than stress in one channel. The result is a fixed 0-100 scale; past values are never revised.
- **No look-ahead.** Every step uses only past data; an automated test checks that adding new data never changes earlier values.





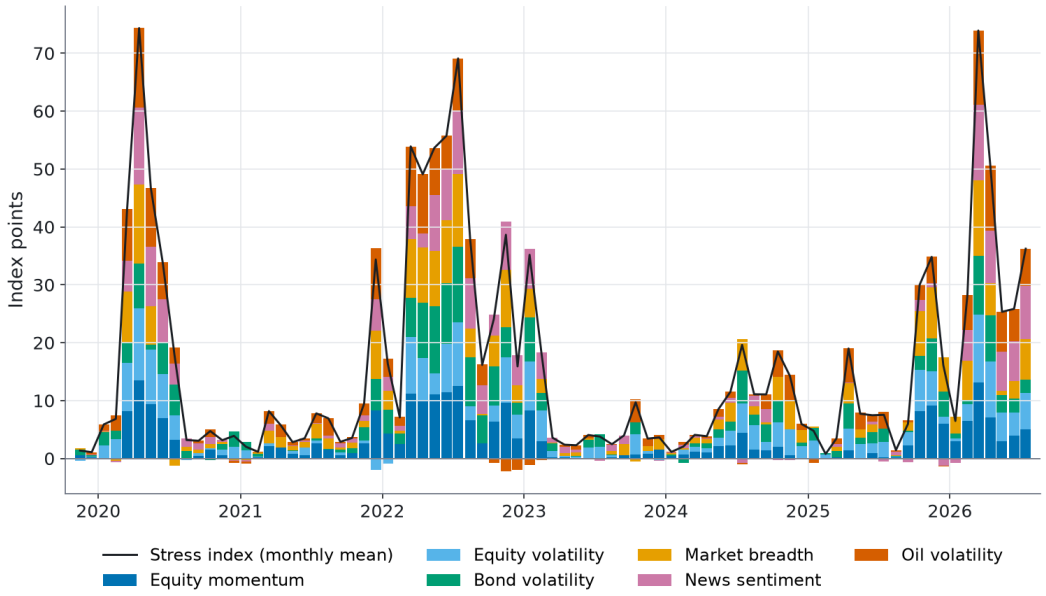
Shading: the 5% most stressed days of the sample (descriptive, uses the full sample).

Results

- **It flags the market's crises.** COVID-19: MASI -28.9% from 22 Jan 2020 to 18 Mar 2020, index peak 80.6; war in Ukraine: MASI -30.5% from 10 Feb 2022 to 6 Jan 2023, index peak 75.3; US tariffs: MASI -10.3% from 24 Mar 2025 to 9 Apr 2025, index peak 27.7. Highest reading: 85.1 on 17 Mar 2026; median 7.6.
- **It separates calm from turbulent markets.** In the calmest fifth of days, MASI volatility over the next 20 trading days averaged 9.4% (annualised), against 13.7% in the most stressed fifth. Correlation with current volatility: 0.57.
- **Real-time labels work in the expected direction.** Days labelled stress were followed by 14.2% volatility, calm days by 11.0% (only 53 stress days).
- **News adds information that prices do not yet show.** Once current volatility is taken into account, the composite keeps little forward information (partial rank correlation 0.06), but the press-sentiment component keeps 0.23.

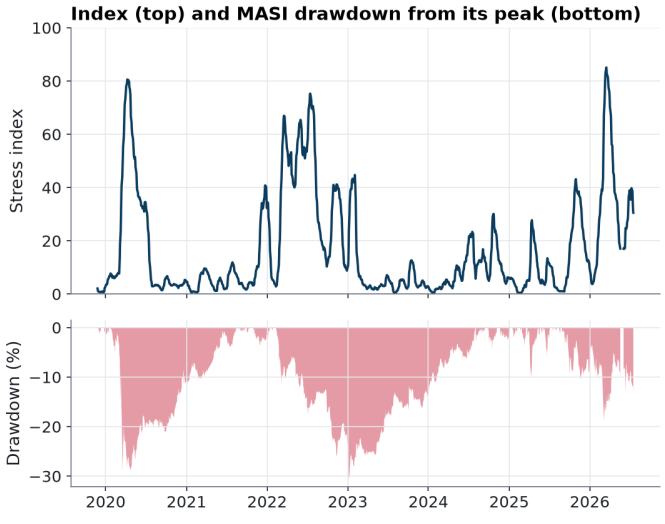
What drives the index

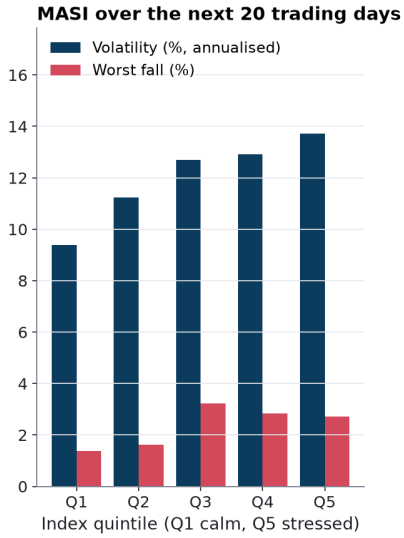
Monthly average contribution of each component, in index points (contributions add up to the index)



Does the index track market stress?

Higher index readings are followed by more volatile markets; the link with future losses is weaker





Limits

- **Not a crash predictor.** The link with losses over the next 20 days is weak (rank correlation -0.12), and the index peaked after the MASI trough in the sharpest crises: it confirms stress more than it anticipates it.
- **In-sample.** Parameters were set on the validation sample. Real-time labels start on 3 Dec 2020, after the COVID-19 shock.
- **Timing.** Day t uses that day's closes and articles; use it from day t+1.
- **Data.** Two news sources and a general-purpose language model; breadth uses the stocks listed today (no delisted firms); days with a missing input are dropped (18 returns span two trading days).